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Safe Isolation Technology for High-Pressure Gas Invasion in Fractured Granite Formation

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Abstract

High-pressure gas formations are extensively distributed in major oilfields. Severe gas invasion encountered during drilling operations may significantly effect well integrity. The conventional plugging method for encountering high-pressure gas is installing packers. However, the plugging success rate and effectiveness are often unsatisfactory. This study optimized the isolation string and operational workflow maximizing isolation capacity and ensuring smooth packer placement, thus offering a solution to enhance open-hole wellbore integrity. To address the challenge of high-pressure gas invasion, a hydraulically expandable external packer was strategically selected. By logging data, the seating depth of the packer has been optimized. To further enhance the string isolation capability, an expandable liner hanger was used to suspend the isolation string. This created a dual - barrier isolation system with expandable liner hanger and packers, establishing favorable wellbore conditions for subsequent production operations. We used an optimized rotary guide shoe to facilitate the smooth deployment of the isolation string in irregular wellbores, which boosts the success rate of such operations. Implementation of this technology enabled the isolation string to reach the designated position and achieve complete seating, attaining a 100% success rate in downhole operations. The application of this technology significantly enhanced the operational efficiency of high-pressure gas isolation, achieving an overlap section reduction of nearly 100 meters compared to conventional hangers. The packer string features an inner diameter exceeding $\varnothing 157\text{mm}$, with a suspension capacity surpassing 50t. It exhibits robust mechanical properties including internal pressure resistance of 45MPa, external collapse resistance of 25MPa, and annular isolation pressure capability of 25MPa. These specifications ensured casing integrity while achieving complete isolation between reservoir layers, thereby establishing a reliable foundation for subsequent zonal testing operations. This technology significantly enhances the isolation capability of high-pressure gas invasion, improves operational efficiency and success rates, while effectively mitigating well control risks. It establishes a solid foundation for subsequent zonal production by optimizing wellbore conditions and provides innovative solutions for enhancing wellbore

integrity in open-hole wells. These advancements demonstrate exemplary implications for the development of analogous oilfields.

Keywords: Well Integrity, Gas invasion, Hydraulically Expandable Packer, Expandable Liner Hanger, Isolation String

Introduction

In the CNPC West Africa B-Basin project, a well-developed fault system overlies a Precambrian fractured-granite reservoir [1]. The granite matrix is extremely hard and abrasive (drillability grade 7), exhibits low pore pressure (0.35–1.00), and has a narrow operational window (0.01–0.02 g/cm³). Consequently, total loss circulation events are common while drilling, often escalating to kicks. Among 39 granite-reservoir wells completed in the past five years, 16 experienced losses and kicks (41.03% of the total). Prolonged subaerial exposure has produced an intensely weathered zone at the granite top. This zone acts as a thief zone that allows high-pressure gas from the overlying formation to channel downward. The resulting open-hole section therefore contains both high-pressure gas and severely depleted zones, creating simultaneous "blow-out below, loss above" scenarios that pose significant operational risks. Conventional mitigation relies on setting external casing packers (ECPs) to isolate the gas. However, irregular borehole diameter frequently prevents the ECP from reaching the planned depth or achieving a competent isolation, resulting in incomplete isolation or outright failure and effect subsequent safe drilling operations.

To address the challenge of high-pressure gas invasion in fractured granite formations, a hydraulically expandable external casing packer (ECP) with high-isolate-pressure was selected. Based on comprehensive interpretation of logging and image log data, the optimal setting depth was determined to enhance isolation success probability. Accounting for liner hanging weight and post-isolation internal diameter constraints, an expandable liner hanger (ELH) was deployed to suspend the isolation string inside the technical casing, thereby creating a redundant "ELH + ECP" barrier system and establishing a robust wellbore architecture for later production operations.

A dedicated clean-out bottom-hole assembly (BHA) and a refined operational sequence were engineered to improve borehole regularity in the granite interval. A rotating guide shoe was installed at the bottom of the isolation string. By combining active and passive rotation mechanisms, the assembly provided precise trajectory control and ensured the string reached the target depth and set successfully, thereby increasing operational reliability.

These technologies constitute a safe, high-pressure gas invasion isolation strategy for fractured granite reservoirs. The integrated approach significantly improves gas isolation capability, enhances operational efficiency and success rate, mitigates well control risks at the same time. It also lays a sound foundation for subsequent zonal production, offers a technical reference for open hole integrity enhancement, and demonstrates demonstrable value for analogous field developments.

Characterization of Fractured Granite Reservoirs

At the base of the B Basin lies a Precambrian granite formation. Superimposed multiphase tectonic movements have induced intensive fracturing, yielding fracture corridors accompanied by vuggy porosity. Prolonged surface exposure produced a pronounced weathering profile, vertically partitioned into four petrophysical zones: (1) the meteoric-leached zone, (2) the fractured zone, (3) the partially cemented zone, and (4) the tight zone. The meteoric-leached zone is dominated by in-situ accumulations of physically disintegrated granitic clasts. Its core analyses yield porosities as high as 22.25 % (average 12.49 %) and permeabilities ranging from 0.36 to 765.06 mD. These attributes render the meteoric-leached zone a primary sweet spot within the granite section [2,3].

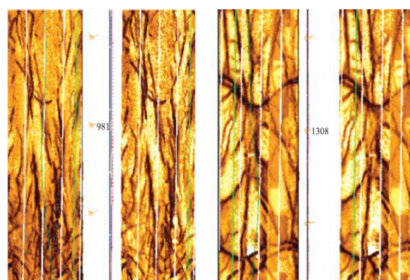


Figure 1—B-2 imaging interpretation results.

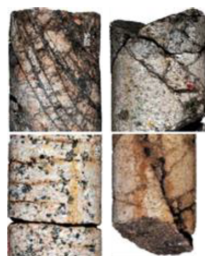


Figure 2—Basement core photographs.

The granite basement is characterized by a hard, dense, and abrasive fabric with poor drillability (grade 7), high Mohs hardness (grade 6), low ductility (grade 2), and elevated unconfined compressive strength (grade 7). Thereby classifying it as a medium-hard, brittle, high-strength rock. During drilling operations, bit damage—including tooth chipping, tooth loss, gauge wear, and bearing failure—occurs frequently. Field performance statistics show an average rate of penetration (ROP) of 2.3 m/h and an average bit run of 171m.

Table 1—Precambrian Granite Drillability Evaluation.

Lithology Type	Test content	Number of Test						mean
		1	2	3	4	5	6	
Precambrian Granite	Hardness / MPa	326	308	326	3.3	168	232	277
	Drillability	6.4	7.1	7.4	6.8	6.8	6.9	6.9
	Plasticity Coefficient	3.16	3.64	2.36	1.33	1.55	2.02	2.34

The fractured granite interval exhibits subnormal pore pressure (pressure coefficient 0.35–1.00) and an extremely narrow drilling-fluid density window (0.01–0.02 g/cm³). Coupled with the presence of open fractures and vugs, these conditions routinely trigger total loss circulation events and, inversely, can induce underbalanced kicks, posing significant well control risks and jeopardizing operational safety.

Of the 39 granite reservoir wells drilled in the last five years, 16 wells experienced losses and kicks, representing 41.03 % of the total. Well Baobab C1-4, for example, registered 13 kicks accompanied by multiple losses, with cumulative mud losses of 624.98 m³. In subsequent granite reservoir drilling, wells Baobab C2-1, Baobab C1-16, and Baobab C1-12 suffered severe losses to the point of complete lost returns at surface; losses amounted to 887.62 m³, 3198.8 m³, and 3229.98 m³, respectively, precluding normal logging operations.

Characteristics and isolation Difficulties of A Formation

Overlying the fractured-granite formation, the A Formation mud-shale interval exhibits porosities of 11–20 % and permeabilities of 10–500 mD, characterizing a medium-porosity, medium-permeability reservoir

that hosts an active gas-bearing zone with abnormal high pressure [4–7]. While drilling Prosopis NE-1, two distinct kicks were taken at 1712m and 1751m, with gas shows rising to 89 % and culminating in surface flow. Baobab S-1 penetrated high pressure gas invasion between 1391m and 1743m; gas readings escalated from 56 % to 100 %, and an influx was observed after pumps were stopped. The well was shut-in and circulated, and the mud weight was increased from 1.20 to 1.46 g/cm³ to regain primary control and reach TD.

The A formation and the granite basement constitute two distinct pressure regimes—pressure coefficients of 0.97–1.15 and 0.35–1.00, respectively—mandating a two-stage drilling program. Current practice sets the casing shoe on top of the granite (minimizing granite exposure to mitigate total-loss risk) and displaces to a solids-free drilling fluid for the subsequent hole section. However, the highly fractured and meteorically leached granite top acts as a thief zone; high pressure gas from the A formation can rapidly migrate downward along open fractures, making reliable casing isolation problematic and leaving the following drilling phase exposed to extreme well-control hazards [8].

Basic Information of Complex Wells

Well O2, a vertical exploratory well located in the B Basin, targeted the fractured granite formation. The well was drilled with a three-casing program:

- $\phi 444.5\text{mm} \times \phi 339.7\text{mm}$ surface hole to 504.8m, setting $\phi 339.7\text{mm}$ casing in stable B formation mudstone to isolate the Q–T and upper B formation sloughing-prone intervals;
- $\phi 311.2\text{mm} \times \phi 244.5\text{mm}$ intermediate hole to 2688.09 m, setting $\phi 244.5\text{mm}$ casing 0.59m into the granite;
- $\phi 215.9\text{mm}$ open hole to TD.

During the $\phi 215.9\text{mm}$ phase, a gas kick was recorded at 2689.5 m. After kill procedures, drilling continued to 2834.94m where total losses occurred. A subsequent influx at 2835.2 m required shut-in, recording a maximum casing pressure of 22.5 MPa. The flare height reached 30 m through the choke. Following LCM squeezes, a bridge plug was set at 2645.5–2646 m and cement was bullheaded from 2595–2645 m to isolate the lower section.

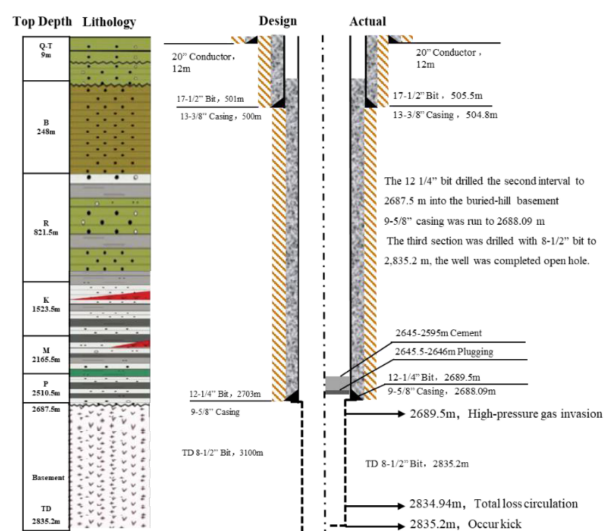


Figure 3—Well Status of O2.

Wireline and LWD interpretations indicate that the granite section remains mechanically competent, with no evidence of massive collapse. The A formation contributes 3.76 m net pay across two zones (Zone 67:

2683.87–2686.12 m), while the granite hosts 17.29 m net pay distributed among 13 discrete layers (Zone 68: 2687.77–2691.41 m). The vertical separation between Zones 67 and 68 is only 1.65 m.

The $\phi 244.5$ mm casing shoe is set at 2688.09 m (1.97 m below the base of A formation Zone 67 and mid-way through granite Zone 68). Cement-bond logs show poor zonal isolation at the shoe.

Four separate test intervals were perforated, spanning the granite, K formation, and R formation; details are provided in the table below.

Oil Test Formation

Num	Formation	Depth(m)	Status
Test#1	Granite Formations	2775.78-2835.2	Openhole
Test#2	Granite Formations	2700.57-2759.63	Openhole
Test#3	K	2038.04-2076.44	Casing
Option#1	R	848.01-849.84	Casing

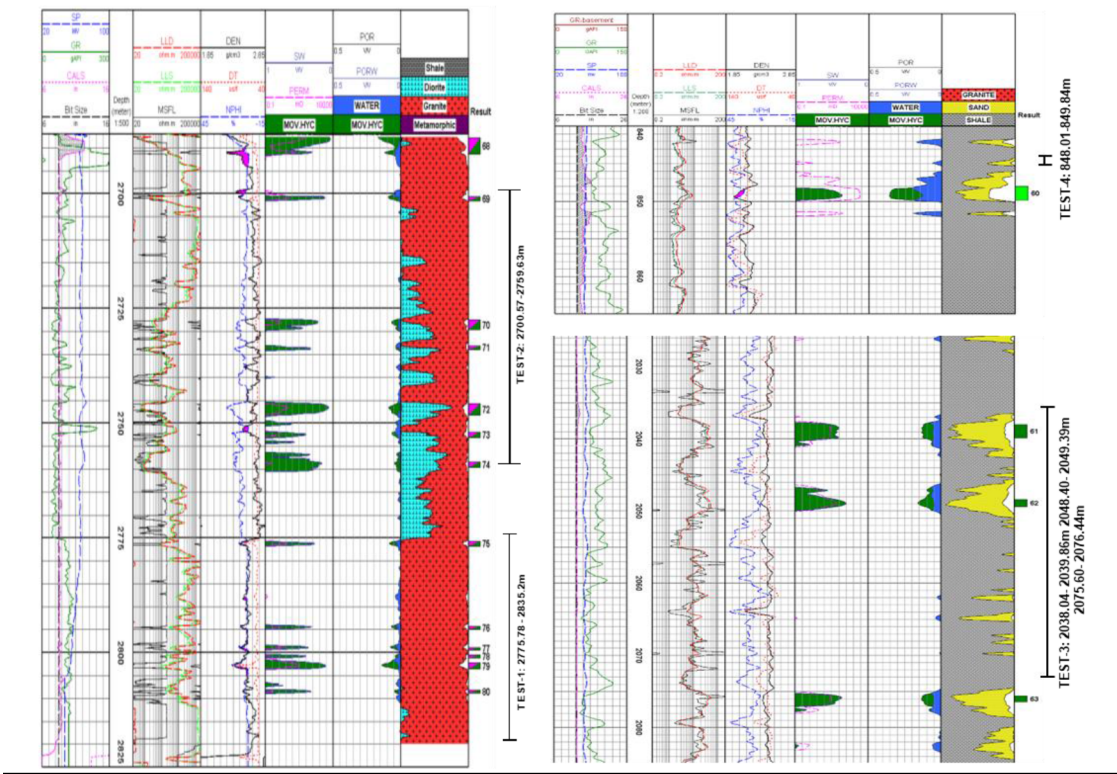


Figure 4—Profile of Mud Logging and Test Interval.

Optimization Design of Isolation String

Expandable Liner Hanger. To tackle the persistent challenge of effectively isolation the high-pressure gas invasion from the lower section of Group A, an expandable liner hanger (ELH) was selected after rigorous screening [9–16]. The hanger delivers high isolation pressure, superior load-carrying capacity and a large drift diameter after expansion. During the operation, the ELH was landed inside the intermediate casing to guarantee a reliable hydraulic insolate of the high-pressure gas invasion interval.

The hanger string comprises a running tool, an expansion cone, the expandable body and elastomeric insulation. The running tool's primary role is to convey the cone, the expandable body and the liner string to the setting depth, and then to trigger radial expansion of the body against the host casing. Expansion is initiated by raising the internal pressure of the running string with surface pumps. This elevated drilling-

fluid pressure acts on the hydraulic chamber inside the running tool, forcing the expansion cone to travel axially through the body. The resulting plastic deformation drives the hanger body outward until it intimately conforms to the inner wall of the last casing. Friction between the expanded body and the casing wall supports the suspended liner string. Simultaneously, the elastomeric rings, pre-installed in external grooves of the hanger body, extrude and expand with the body, occupying the micro-annulus between the hanger OD and the casing ID. This dual action creates a high-integrity annular insulate while augmenting the frictional grip required for long-term mechanical anchorage.

The expandable liner hanger (ELH) integrates metallic, elastomeric and composite barriers to isolate the annulus at pressures exceeding 25 MPa, providing a robust safeguard against gas invasion. Its axial load rating surpasses 90 t, readily accommodating the weight of extended open-hole isolation strings encountered in the field. Post-expansion, the hanger preserves a drift diameter greater than $\phi 190$ mm, ensuring unimpeded passage for subsequent completion assemblies and artificial-lift pumps. The minimal step-out to the parent casing, complemented by a V-shaped bell-mouth entry, markedly reduces the risk of differential sticking or mechanical obstruction. By permitting an overlap as short as 30 m, the tool also delivers appreciable savings in rig time and material cost.

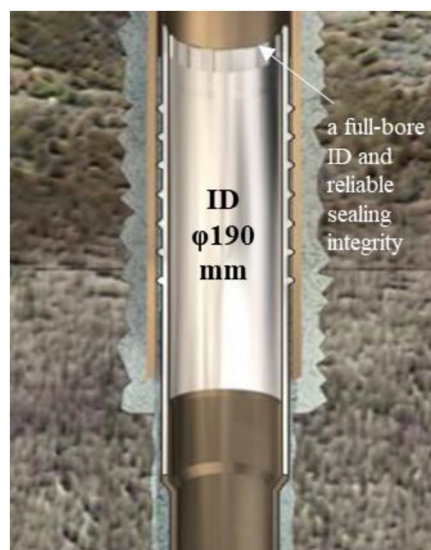


Figure 5—Expandable Liner Hanger.

Hydraulically Expandable Packer. To address the challenge of severe gas invasion in Well O2 and the need to conduct formation testing across multiple intervals, a hydraulically expandable external casing packer was selected [17–19]. This tool, a hydraulically inflated downhole device, is specifically designed for complex oil, gas and water well completions to achieve downhole isolation and zonal plugging. After the packer is run to the target depth, hydraulic pressure is applied through the tubing; once the preset pressure is reached, the element expands and anchors against the borehole wall to establish isolation.

Key performance attributes include a high expansion ratio, a 30MPa rated isolation pressure, reliable set-and-release characteristics, elevated temperature tolerance, and a design life exceeding five years. An internal drift of $\phi 157$ mm is maintained after setting, and the 30MPa isolation capability ensures effective containment of upward invasion high pressure gas. The packer can be deployed in the tight streaks separating productive layers, enabling discrete, multi-stage testing without intervention.

Currently, the $\phi 244.5$ mm casing shoe is set 1.97 m above the bottom of Reservoir No. 67 and lies within the granite Reservoir No. 68. Cement-bond log indicates poor cement integrity at the shoe.

Image-log interpretation confirms that the inter-layer intervals are non-reservoir, tight formations, satisfying the setting requirements for an external-casing packer (ECP). Caliper data were extracted for

four depth ranges—between Reservoirs 68 and 69, 74 and 75, and from Reservoir 80 to total depth (TD). After ranking the borehole regularity, three intervals exhibiting the most uniform gauge were selected as candidate ECP setting depths:

2692–2699 m: maximum diameter 233.19 mm, minimum 215.85 mm, average 219.42 mm.

2760–2775 m: maximum diameter 226.81 mm, minimum 215.26 mm, average 219.83 mm.

2810–2818 m: maximum diameter 229.05 mm, minimum 215.26 mm, average 223.84 mm.

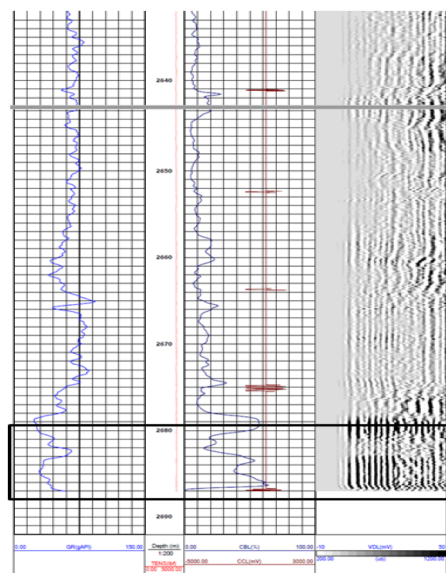


Figure 6—Casing Shoe Cementing Quality.

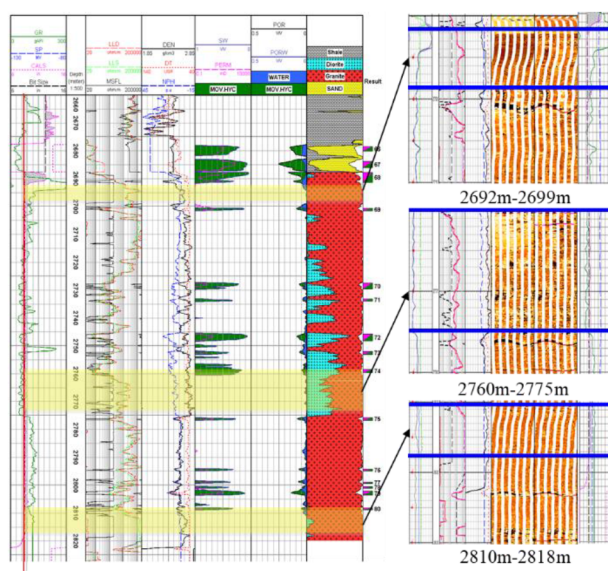


Figure 7—Well Diameter, Rock Properties and Imaging Logging Analysis.

In alignment with the planned testing intervals, three external-casing packers (ECPs) were deployed to isolate (1) the shoe to top of Reservoir 69 section, (2) the base of Reservoir 74 to top of Reservoir 75 section, and (3) the base of Reservoir 80 to TD section. This configuration simultaneously shuts off the gas invasion interval and permits staged testing of the granite reservoirs. After isolation, an "APR-TCP" combination test string was run to evaluate Test #1 and Test #2 (perforating guns positioned in the $\phi 177.8$ mm casing interval, with an RTTS packer set in the $\phi 244.5$ mm casing). Upon completion, a composite bridge plug plus

cement plug was spotted to isolate the lower zones, the well was then sidetracked back to the K/R interval, and a second "APR-TCP" string was deployed for testing the K/R formation.

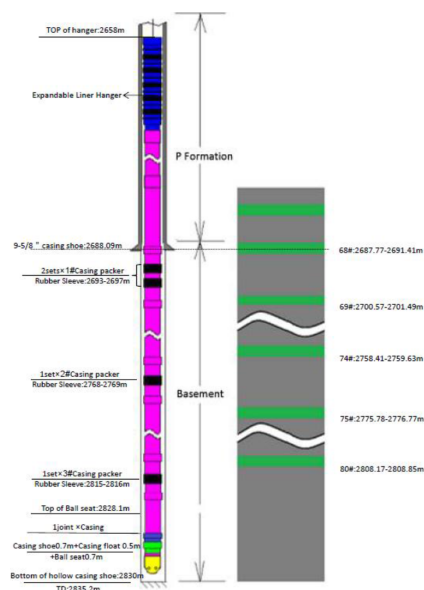


Figure 8—Isolation String.

Rotary Guide Shoe. The granite formation exhibits high rock strength, limiting the conventional reaming BHA's ability to dress the borehole wall and irregular hole geometry in turn jeopardizes the safe run-in of the isolation string. If the packer fails to set at target depth, the high-pressure gas cannot be effectively isolated, compromising subsequent safe operations.

To improve packer running success, shorten operation time and mitigate well control risk, a self-rotating guide shoe (SRS) was selected [20–25]. Positioned at the bottom of string, the SRS integrates mechanical self-orientation and hydraulic rotation functions. When casing encounters obstruction, drilling fluid pumped through the SRS central port actuates the internal turbine, driving the transmission shaft and eccentric reaming head. Tungsten-carbide blades and circulation nozzles on the reaming head trim micro-ledges to enhance borehole clearance while fluid jets evacuate cuttings, ensuring one-trip casing success.

Following successful casing running to the target depth, increased flow rate shears the pins, driving the turbine shaft downward to isolate the flow path. When annular pressure exceeds the predetermined value, the rupture disk fails, establishing a new flow path for subsequent cement slurry placement. Upon completion of cementing operations, the assembly can be drilled out using a roller cone or PDC bit to enable drilling of the subsequent section.

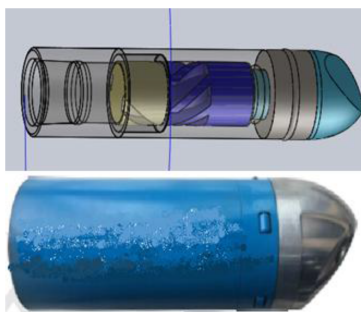


Figure 9—Rotary Guide Shoe.

Isolation String Optimization. Based on integrated analyses of drilling, logging, and LWD data, the isolation string was optimized by incorporating an expandable liner hanger, a hydraulically activated external casing packer (ECP), and a rotating guide shoe tool. The final configuration is rotating guide shoe + 7" float collar + 7" ball seat + 7" casing + two 7" ECPs + 7" casing + two 7" ECPs + 7" casing + three 7". ECPs + 7" casing + expandable liner hanger + sub + 3-1/2" drill pipe. The hanger is set at 2,658 m MD, the isolation string bottom is at 2,830 m MD, resulting in an isolation length of approximately 172 m with a 30m overlap against the previous casing string, and the ball seat is landed directly above the float collar. Circulation capability was retained at any point along the string to enable spotting LCM or kill-weight mud in the event of losses or gas invasion. A back-pressure valve was integrated to provide an additional barrier against well control events.

The liner hanger annulus isolation is rated to 25 MPa and delivers a permanent, metal-elastomer barrier between the 9-5/8" host casing and the 7" liner. The expandable liner hanger delivers a large drift diameter (ID > 7-1/2") and forms only a minimal step against the upper casing. Its V-shaped bell-mouth entry further minimizes sticking risk. Three external casing packers isolate the open-hole interval. Each packer is rated to 30 MPa, ensuring element-to-formation contact and isolate the high-pressure gas invasion. Tool ID is $\phi 157$ mm with a service life exceeding five years. When stuck happen in casing running, circulate mud to spin the rotary shoe turbine. The shaft turns the drive sub and eccentric guide nose, steering the string past ledges to depth.

After successful isolation, an "APR + TCP" combined testing string was run to flow-test the Test#1 and Test#2 intervals. A bridge plug plus cement plug was then set to isolate these zones, and the same test string was rerun to evaluate the K/R interval.

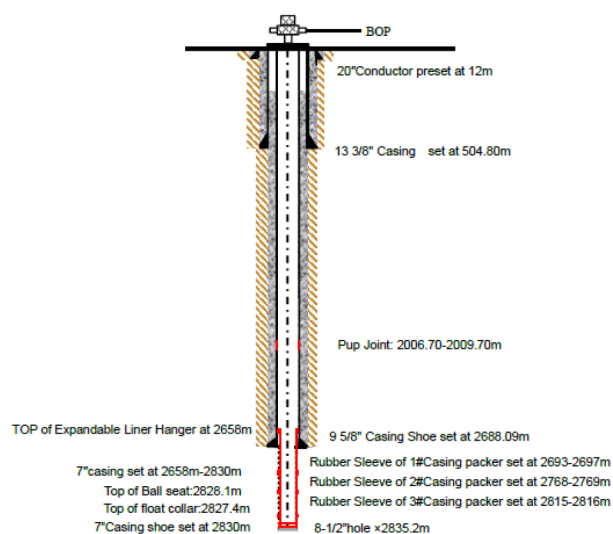


Figure 10—Well Status After Isolation.

Operation Workflow Optimization

The isolation string operation workflow optimization for Well O2 was streamlined into four critical stages: casing clean-out, drift run, wireline depth correlation, and pressure-set activation. A dual-depth-check protocol was implemented: Run-1, executed immediately after the drift run, verifies that the open-hole section is mechanically acceptable for the completion string; Run-2, performed after the isolation string is landed, confirms that the packer train has reached the target depth. Once confirmed, applied surface pressure activates the expander cone, simultaneously setting the liner hanger and energizing the ECP elements.

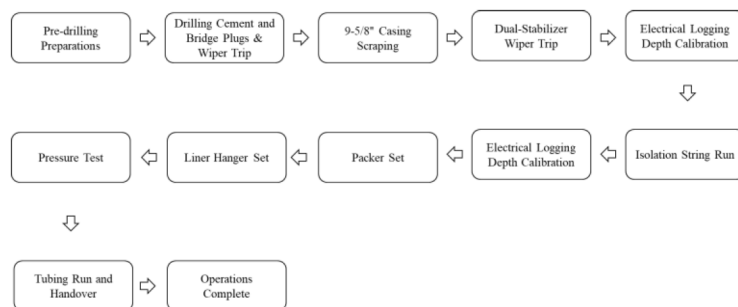


Figure 11—Operation Workflow Optimization.

The clean-out string was optimized for this wellbore as follows: 9-5/8" casing scraper+ 6-1/2" float valve + 8-3/8" stabilizer+ eight joints of 6-1/2" drill collars+ 12 joints of 5" heavy weight drill pipe+ 5" drill pipe.

To counter the high competence and abrasiveness of the granite, a dual-stabilizer drift BHA was engineered to maximize borehole regularity and ensure trouble-free isolation string deployment. A Barat impregnated bit was selected for its thermally-stable, ultra-abrasive-resistant cutting structure that shears and grinds the formation, delivering superior wall-conditioning in granite. Full-gauge rigid stabilizers were spaced to maintain gauge and smooth micro-ledges.

BHA configuration: 8-1/2" Barat PDC bit+ 6-1/2" float sub+ 8-1/2" full-gauge rigid stabilizer+ one joint 6-1/2" drill collar+ two 8-3/8" rigid stabilizers+ eight joints 6-1/2" drill collars+ fifteen joints 5" heavy weight drill pipe+ 5" drill pipe.

Application Effect

After completing the pre-job casing scraping and drift runs, the isolation string was run in hole for Well O2. Once the string tagged bottom, an electric-line correlation log was performed and confirmed the assembly had landed at the programmed depth. The annulus packer and the expandable liner hanger were subsequently pressure-set and successfully deployed. The ensuing full-bore pressure test met the isolation verification specification. By integrating the tailored "expandable liner hanger + hydraulically inflated external casing packer + rotating guide shoe" technique, the high-pressure gas invasion in Well O2 was safely isolated, securing the subsequent staged testing operations.



Figure 12—Packer Running into the Well.

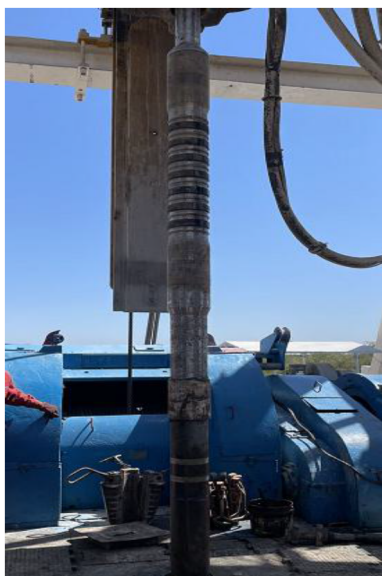


Figure 13—Hanger Running into the Well.

Conclusions

1. The invasion of high-pressure gas into the open-hole section severely effects the safety of subsequent zonal formation testing, placing more stringent demands on the gas-tight integrity of the isolation tools. Deployment of a dual-isolation system (combining an expandable liner hanger with a packer) maximized isolation capacity and enhanced the isolation performance against high-pressure gas invasion. The technology reduces the overlap section by nearly 100 m compared with a conventional hanger. The isolation string maintains an ID above $\phi 157$ mm and delivers a hanging capacity exceeding 50 t. It withstands 45 MPa internal pressure and 25 MPa collapse. The annular seal rating reaches 25 MPa. These specifications safeguard wellbore integrity and enable complete zonal isolation between pay intervals.
2. A self-rotating guide shoe at the string bottom mitigates sticking risk. A drift BHA utilizing a Barat bit and dual full-gauge rigid stabilizers was engineered to optimize borehole quality, enabling the isolation string to reach TD in one run.
3. Considering the formation characteristics and contingencies, the workflow was optimized, boosting operational efficiency and delivering a 100 % success rate.
4. Field execution in Well O2 confirmed that the Expandable Liner Hanger+ Hydraulically Expandable Packer+ Rotary Guide Shoe string is an effective solution for high pressure gas invasion isolation, providing a replicable template for analogous wells and ensuring safe and efficient production operations.

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